

Week 2 Day 12 Vision Data Review

Name:- Shivam Kumar

Group:- C2

Dataset V1 – EDA and Pairing

Dataset Statistics

- Input images: 1078
- Target images: 1078
- Generated images: 1078
- Image size: 256 × 256
- Image format: PNG

Difficult Conditions

- Very dark images
- Low-contrast images
- Different illumination levels

EDA Findings

- Average input brightness: 70.16
- Minimum brightness: 1.03
- Maximum brightness: 246.69
- Average input contrast: 26.68
- Minimum contrast: 2.19
- Maximum contrast: 73.32

Pairing Finding

- A duplicate file was identified:
- img 74_input - Copy.png
- It was confirmed as an exact duplicate.
- Duplicate samples will be excluded from the final dataset.

Preprocessing and Dataset Split Strategy

Preprocessing

<u>Step</u>	<u>Decision</u>
Image format	PNG
Color	RGB
Image size	256 × 256
Normalization	0–1
Input	Underwater image
Target	Clear/reference image

Augmentation

No augmentation for V1

Reason:

- This is an image-to-image enhancement task.
- Input and target images must remain spatially aligned.
- Unnecessary transformations could change the image degradation relationship.

Split Logic

Valid Input-Target Pairs



Shuffle (seed=42)



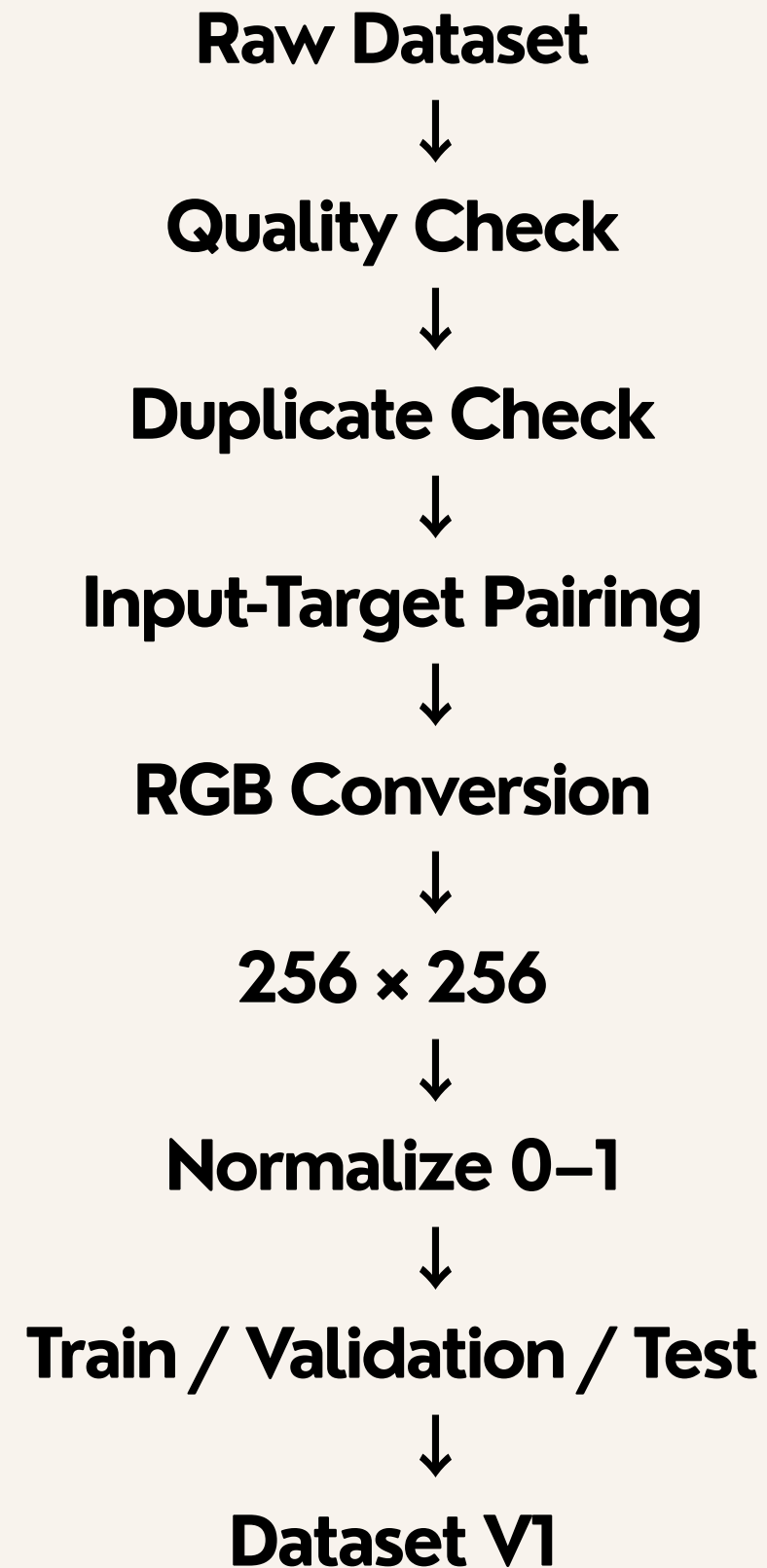
70% Train

15% Validation

15% Test

Dataset V1 – Final Preparation Plan

Final Workflow



Configuration

- Version: V1
- Image size: 256 × 256
- Channels: 3
- Normalization: 0–1
- Batch size: 8
- Random seed: 42
- Augmentation: None

Dataset V1

<u>Split</u>	<u>Pairs</u>
Train	218
Validation	47
Test	47
Total	312



THANK YOU!

By Olivia Wilson